CNC Machine Troubleshooting Manual

A practical guide to diagnosing and fixing common issues in six types of CNC machines.

1. CNC Milling Machine

- Spindle not rotating → Check spindle motor power supply, inspect spindle drive, reset overload relay.
- Poor surface finish → Replace dull tools, tighten fixtures, adjust spindle speed and feed rate.
- Axis not moving → Inspect servo motors/drives, check ball screw alignment, verify lubrication system.

2. CNC Lathe

- Workpiece chatter → Tighten workholding (chuck/collet), reduce tool overhang, increase rigidity.
- Spindle vibration → Inspect bearings, balance chuck and workpiece, check tailstock alignment.
- Tool wear too fast → Verify coolant flow, use correct cutting parameters, choose appropriate tool material.

3. CNC Router

- Machine loses steps → Check drivers, tighten belts and pulleys, reduce feed rate.
- Burn marks on material → Increase feed rate, ensure sharp cutting tool, check spindle RPM.
- Dust extraction issues → Inspect vacuum system, clean filters, ensure proper airflow.

4. CNC Plasma Cutter

- Poor cut quality (slag, dross) → Adjust cutting speed, check plasma gas pressure, replace worn consumables.
- Arc failing to start → Verify grounding clamp, check air pressure, replace pilot arc components.
- Irregular kerf width → Inspect torch alignment, replace nozzle, check cutting height control.

5. CNC Laser Cutter

- Not cutting through material → Clean or replace lens/mirrors, adjust focus height, verify power settings.
- Burn marks on edges → Increase cutting speed, ensure assist gas flow, clean optics.
- Misaligned cuts → Recalibrate machine axes, check belts or pulleys, ensure workpiece is fixed securely.

6. CNC EDM Machine

- Excessive electrode wear → Use correct polarity, adjust flushing conditions, optimize current settings.
- Slow machining speed → Increase discharge current, improve dielectric fluid flushing, replace electrodes.
- Arcing & short circuits → Check dielectric fluid contamination, verify flushing nozzles, inspect alignment.

General Preventive Maintenance (For All CNC Machines)

- Clean chips, dust, and residues after every operation.
- Check lubrication and coolant levels daily.
- Inspect spindle bearings, belts, and ball screws weekly.
- Calibrate axes monthly.
- Replace worn consumables (cutting tools, nozzles, electrodes) on schedule.
- Keep electrical cabinets dust-free.

Safety Guidelines

- Always wear safety goggles, gloves, and protective clothing.
- Keep tools and machines away from unauthorized personnel.
- Disconnect power supply before maintenance or repair.
- Do not use damaged or malfunctioning machines until fixed.
- Ensure proper ventilation when operating plasma or laser cutters.

Conclusion

This troubleshooting manual covers common problems and fixes for six types of CNC machines. By following these guidelines, operators and technicians can improve machine reliability, minimize downtime, and maintain safe and efficient operations.